NEWHAVEN DISPLAY INTERNATIONAL

**Product Specification**

# NHD-0440AZ-RN-FBW

## Character Liquid Crystal Display Module

| | |
|---|---|
| **NHD-** | Newhaven Display |
| **0440-** | 4 Lines x 40 Characters |
| **AZ-** | Model |
| **R-** | Reflective |
| **N-** | No Backlight |
| **F-** | FSTN (+) |
| **B-** | 6:00 Optimal View |
| **W-** | Wide Temperature |

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REACH Compliant

RoHS Compliant

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## Additional Resources

- **Support Forum:** https://support.newhavendisplay.com/hc/en-us/community/topics
- **GitHub:** https://github.com/newhavendisplay
- **Example Code:** https://support.newhavendisplay.com/hc/en-us/categories/4409527834135-Example-Code/
- **Knowledge Center:** https://www.newhavendisplay.com/knowledge_center.html
- **Quality Center:** https://www.newhavendisplay.com/quality_center.html
- **Precautions for using LCDs/LCMs:** https://www.newhavendisplay.com/specs/precautions.pdf
- **Warranty / Terms & Conditions:** https://www.newhavendisplay.com/terms.html

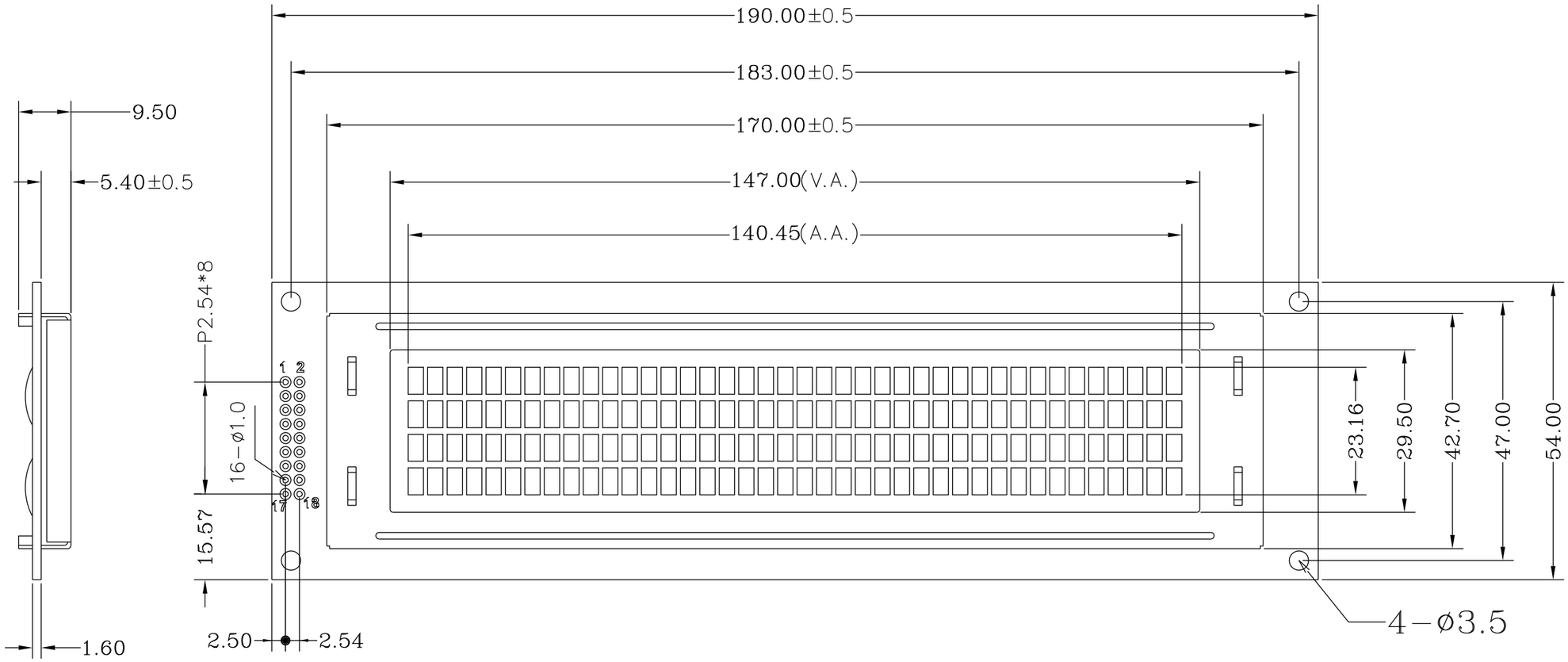
# Document Revision History

| Revision | Date | Description | Changed By |
|---|---|---|---|
| 0 | 10/21/2008 | Initial Release | - |
| 1 | 11/02/2009 | User Guide Reformat | MC |
| 2 | 11/16/2009 | Updated Block diagram and Initialization Code | MC |
| 3 | 01/05/2011 | Updating Driver Information | JT |
| 4 | 05/06/2011 | Electrical Characteristics Updated | AK |
| 5 | 11/21/2011 | Mechanical Drawing Updated | TJ |
| 6 | 01/11/2026 | Document Formatting Updated | KL |

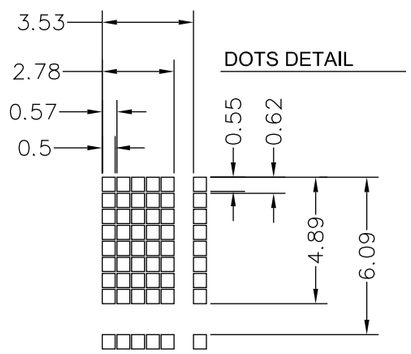
# Mechanical Drawing

Newhaven Display
NHD-0440AZ-RN-FBW
Date Code

Part Label (type/format may vary)




| | |
|---|---|
| 1 | DB7 |
| 2 | DB6 |
| 3 | DB5 |
| 4 | DB4 |
| 5 | DB3 |
| 6 | DB2 |
| 7 | DB1 |
| 8 | DB0 |
| 9 | E1 |
| 10 | RW |
| 11 | RS |
| 12 | V0 |
| 13 | VSS |
| 14 | VDD |
| 15 | E2 |
| 16 | NC |
| 17 | NC |
| 18 | NC |




Product Description: 4x40 Character LCD

1. Driver IC: ST7066U
2. Driving Mode: 1/16 Duty, 1/5 Bias
3. Interface: 4-bit 6800 Parallel
4. Power Requirement: 5.0V LCD
5. Optical Features: FSTN (+), Reflective, 6:00 View
6. Recommended Pin Header: 2x9pin 2.54mm pitch

**Standard Tolerance:**
(Unless otherwise specified)
Linear: ±0.3mm

NEWHAVEN DISPLAY INTERNATIONAL

Drawing/Part Number: NHD-0440AZ-RN-FBW
Revision: -

**Unless otherwise specified:**
• Dimensions are in Millimeters
• Third Angle Projection

Drawn By: K. Lewis
Approved By: K. Lewis
Drawn Date: 01/11/2026
Approved Date: 01/11/2026

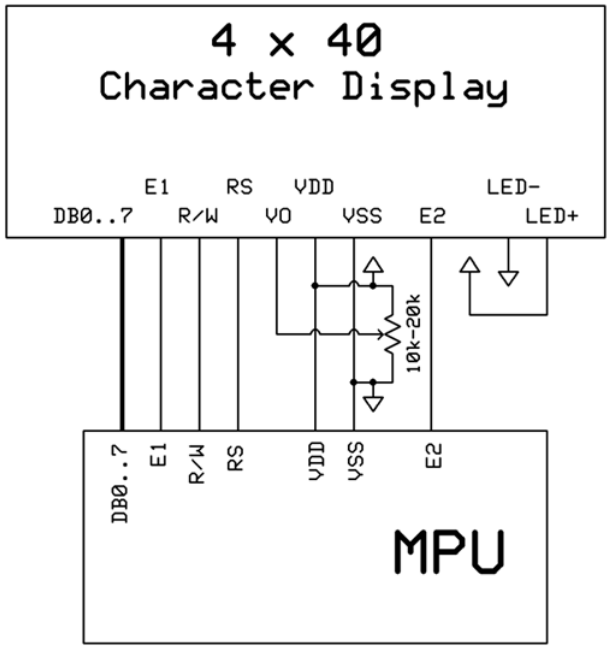
This drawing is solely the property of Newhaven Display International, Inc. The information it contains is not to be disclosed, reproduced or copied in whole or part without written approval from Newhaven Display.

# Pin Description

| Pin No. | Symbol | External Connection | Function Description |
|---|---|---|---|
| 1-4 | DB7-DB4 | MPU | Four high order bi-directional three-state data bus lines. |
| 5-8 | DB3-DB0 | MPU | Four low order bi-directional three-state data bus lines. These four are not used during 4-bit operation. |
| 9 | E1 | MPU | Operation enable signal. Falling edge triggered for top 2 lines. |
| 10 | R/W | MPU | Read/Write select signal, R/W=1: Read R/W:=0: Write |
| 11 | RS | MPU | Register select signal. RS=0: Command, RS=1: Data |
| 12 | V0 | Power Supply | Power supply for contrast (approx. +0.5V) |
| 13 | Vss | Power Supply | Ground |
| 14 | VDD | Power Supply | Supply voltage for logic (+5.0V) |
| 15 | E2 | MPU | Operation enable signal. Falling edge triggered for bottom 2 lines. |
| 16 | NC | - | No Connect |
| 17 | NC | - | No Connect |
| 18 | NC | - | No Connect |

**Recommended LCD connector:** 2.54mm pitch pins
**Backlight connector:** ---  **Mates with:** ---

# Wiring Diagram




# DDRAM Address

| 1 | 2 | 3 | 4 | 5 | - | - | - | - | - | - | - | - | - | - | 36 | 37 | 38 | 39 | 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 00 | 01 | 02 | 03 | 04 | - | - | - | - | - | - | - | - | - | - | 23 | 24 | 25 | 26 | 27 |
| 40 | 41 | 42 | 43 | 44 | - | - | - | - | - | - | - | - | - | - | 63 | 64 | 65 | 66 | 67 |
| 00 | 01 | 02 | 03 | 04 | - | - | - | - | - | - | - | - | - | - | 23 | 24 | 25 | 26 | 27 |
| 40 | 41 | 42 | 43 | 44 | - | - | - | - | - | - | - | - | - | - | 63 | 64 | 65 | 66 | 67 |

# Pin Description

| Item | Symbol | Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| Operating Temperature Range | $T_{OP}$ | Absolute Max | -20 | - | +70 | °C |
| Storage Temperature Range | $T_{ST}$ | Absolute Max | -30 | - | +80 | °C |
| Supply Voltage | $V_{DD}$ | | 4.7 | 5.0 | 5.5 | V |
| Supply Current | $I_{DD}$ | Ta=25°C, VDD=5.0V | - | 3.5 | 4.0 | mA |
| Supply for LCD (contrast) | VDD-V0 | Ta=25°C | - | 4.5 | - | V |
| "H" Level input | $V_{IH}$ | - | 0.7 VDD | - | VDD | V |
| "L" Level input | $V_{IL}$ | - | 0 | - | 0.6 | V |
| "H" Level output | $V_{OH}$ | - | 3.9 | - | - | V |
| "L" Level output | $V_{OL}$ | - | - | - | 0.4 | V |
| | | | | | | |
| Backlight Supply Voltage | $V_{LED}$ | - | - | - | - | V |
| Backlight Supply Current | $I_{LED}$ | - | - | - | - | mA |

# Optical Characteristics

| Item | | Symbol | Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| Optimal Viewing Angles | Top | φY+ | CR ≥ 3 | - | 20 | - | ° |
| | Bottom | φY- | | - | 50 | - | ° |
| | Left | θX- | | - | 30 | - | ° |
| | Right | θX+ | | - | 30 | - | ° |
| Contrast Ratio | | CR | - | 3 | 5 | - | - |
| Response Time | Rise | $T_R$ | $T_{OP}$ = 25°C | - | 150 | 250 | ms |
| | Fall | $T_F$ | | - | 150 | 250 | ms |

# Controller Information

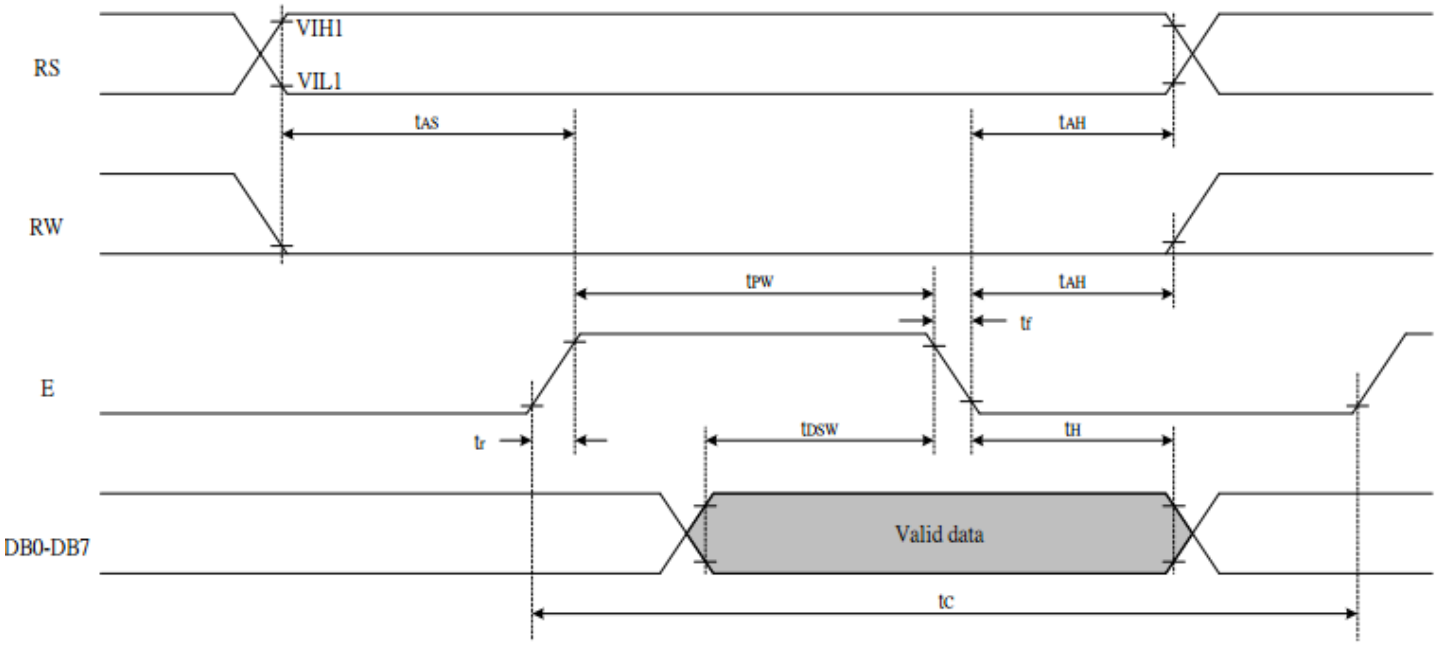
Built-in ST7066U Controller: https://support.newhavendisplay.com/hc/en-us/articles/4414848155159-ST7066U

# Table of Commands

| Instruction | Instruction Code | | | | | | | | | | Description | Description Time (270KHz) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | RS | R/W | DB7 | DB6 | DB5 | DB4 | DB3 | DB2 | DB1 | DB0 | | |
| Clear Display | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | Write "20H" to DDRAM. and set DDRAM address to "00H" from AC | 1.52 ms |
| Return Home | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | x | Set DDRAM address to "00H" from AC and return cursor to its original position if shifted. The contents of DDRAM are not changed. | 1.52 ms |
| Entry Mode Set | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | I/D | S | Sets cursor move direction and specifies display shift. These operations are performed during data write and read. | 37 us |
| Display ON/OFF | 0 | 0 | 0 | 0 | 0 | 0 | 1 | D | C | B | D=1:entire display on<br>C=1:cursor on<br>B=1:cursor position on | 37 us |
| Cursor or Display Shift | 0 | 0 | 0 | 0 | 0 | 1 | S/C | R/L | x | x | Set cursor moving and display shift control bit, and the direction, without changing DDRAM data. | 37 us |
| Function Set | 0 | 0 | 0 | 0 | 1 | DL | N | F | x | x | DL:interface data is 8/4 bits<br>N:number of line is 2/1<br>F:font size is 5x11/5x8 | 37 us |
| Set CGRAM address | 0 | 0 | 0 | 1 | AC5 | AC4 | AC3 | AC2 | AC1 | AC0 | Set CGRAM address in address counter | 37 us |
| Set DDRAM address | 0 | 0 | 1 | AC6 | AC5 | AC4 | AC3 | AC2 | AC1 | AC0 | Set DDRAM address in address counter | 37 us |
| Read Busy flag and address | 0 | 1 | BF | AC6 | AC5 | AC4 | AC3 | AC2 | AC1 | AC0 | Whether during internal operation or not can be known by reading BF. The contents of address counter can also be read. | 0 us |
| Write data to RAM | 1 | 0 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Write data into internal RAM (DDRAM/CGRAM) | 37 us |
| Read data from RAM | 1 | 1 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Read data from internal RAM (DDRAM/CGRAM) | 37 us |

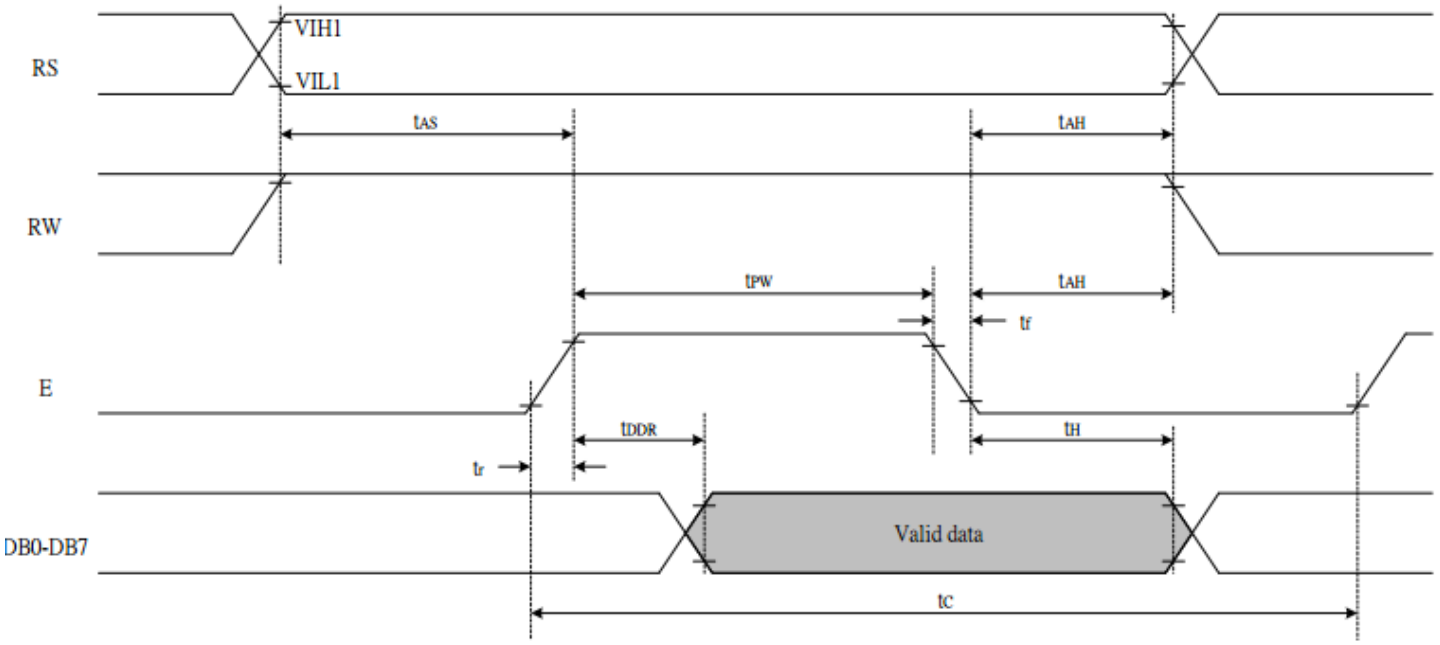
# Timing Characteristics

### Writing data from MPU to ST7066U




| *Write Mode (Writing data from MPU to ST7066U)* | | | | | | |
|---|---|---|---|---|---|---|
| $T_C$ | Enable Cycle Time | Pin E | 1200 | - | - | ns |
| $T_{PW}$ | Enable Pulse Width | Pin E | 140 | - | - | ns |
| $T_R, T_F$ | Enable Rise/Fall Time | Pin E | - | - | 25 | ns |
| $T_{AS}$ | Address Setup Time | Pins: RS,RW,E | 0 | - | - | ns |
| $T_{AH}$ | Address Hold Time | Pins: RS,RW,E | 10 | - | - | ns |
| $T_{DSW}$ | Data Setup Time | Pins: DB0 - DB7 | 40 | - | - | ns |
| $T_H$ | Data Hold Time | Pins: DB0 - DB7 | 10 | - | - | ns |

**Reading data from ST7066U to MPU**




| *Read Mode (Reading Data from ST7066U to MPU)* | | | | | | |
|---|---|---|---|---|---|---|
| $T_C$ | Enable Cycle Time | Pin E | 1200 | - | - | ns |
| $T_{PW}$ | Enable Pulse Width | Pin E | 140 | - | - | ns |
| $T_R, T_F$ | Enable Rise/Fall Time | Pin E | - | - | 25 | ns |
| $T_{AS}$ | Address Setup Time | Pins: RS,RW,E | 0 | - | - | ns |
| $T_{AH}$ | Address Hold Time | Pins: RS,RW,E | 10 | - | - | ns |
| $T_{DDR}$ | Data Setup Time | Pins: DB0 - DB7 | - | - | 100 | ns |
| $T_H$ | Data Hold Time | Pins: DB0 - DB7 | 10 | - | - | ns |

## Built-in Font table

| Lower 4 Bits \ Upper 4 Bits | 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| xxxx0000 | CG RAM (1) | | | 0 | @ | P | ` | p | | | | ー | タ | ミ | α | p |
| xxxx0001 | (2) | | ! | 1 | A | Q | a | q | | | 。 | ア | チ | ム | ä | q |
| xxxx0010 | (3) | | " | 2 | B | R | b | r | | | 「 | イ | ツ | メ | ß | θ |
| xxxx0011 | (4) | | # | 3 | C | S | c | s | | | 」 | ウ | テ | モ | ε | ∞ |
| xxxx0100 | (5) | | $ | 4 | D | T | d | t | | | 、 | エ | ト | ヤ | μ | Ω |
| xxxx0101 | (6) | | % | 5 | E | U | e | u | | | ・ | オ | ナ | ユ | σ | ü |
| xxxx0110 | (7) | | & | 6 | F | V | f | v | | | ヲ | カ | ニ | ヨ | ρ | Σ |
| xxxx0111 | (8) | | ' | 7 | G | W | g | w | | | ァ | キ | ヌ | ラ | g | π |
| xxxx1000 | (1) | | ( | 8 | H | X | h | x | | | ィ | ク | ネ | リ | √ | x̄ |
| xxxx1001 | (2) | | ) | 9 | I | Y | i | y | | | ゥ | ケ | ノ | ル | -¹ | y |
| xxxx1010 | (3) | | * | : | J | Z | j | z | | | ェ | コ | ハ | レ | j | 千 |
| xxxx1011 | (4) | | + | ; | K | [ | k | { | | | ォ | サ | ヒ | ロ | ˣ | 万 |
| xxxx1100 | (5) | | , | < | L | ¥ | l | \| | | | ャ | シ | フ | ワ | ¢ | 円 |
| xxxx1101 | (6) | | − | = | M | ] | m | } | | | ュ | ス | ヘ | ン | ŧ | ÷ |
| xxxx1110 | (7) | | . | > | N | ^ | n | → | | | ョ | セ | ホ | ゛ | ñ | |
| xxxx1111 | (8) | | / | ? | O | _ | o | ← | | | ッ | ソ | マ | ° | ö | ■ |

# Example Initialization Program

```
/*************************************************************************/
void command1(char i)                    //Top half of the display
{
        P1 = i;
        W = 0;
        RS = 0;
        E1 = 1;
        delay(2);
        E1 = 0;
}
void command2(char i)                    //Bottom half of the display
{
        P1 = i;
        W = 0;
        RS = 0;
        E2 = 1;
        delay(2);
        E2 = 0;
}
/*************************************************************************/
void writedata1(char i)                  //Top half of the display
{
        P1 = i;
        W = 0;
        RS = 1;
        E1 = 1;
        delay(2);
        E1 = 0;
}
void writedata2(char i)                  //Bottom half of the display
{
        P1 = i;
        W = 0;
        RS = 1;
        E2 = 1;
        delay(2);
```

```
        E2 = 0;
}
/*****************************************************************************/
void init_LCD()
{
        delay(15);
        command1(0x30);         //Wake up
        command2(0x30);
        delay(5);
        command1(0x30);         //Wake up
        command2(0x30);
        delay(5);
        command1(0x30);         //Wake up
        command2(0x30);
        delay(5);
        command1(0x38);         //Function Set = 8bit mode; 2-line; 5x8
        command2(0x38);
        command1(0x08);         //Turn off display
        command2(0x08);
        command1(0x01);         //Clear display
        command2(0x01);
        command1(0x06);         //Entry mode cursor increment
        command2(0x06);
        command1(0x0c);          //Turn on display; no cursor
        command2(0x0c);
}
/*****************************************************************************/
```

# Quality Information

| Test Item | Content of Test | Test Condition | Note |
|---|---|---|---|
| High Temperature storage | Endurance test applying the high storage temperature for a long time. | +80°C , 48hrs | 2 |
| Low Temperature storage | Endurance test applying the low storage temperature for a long time. | -30°C , 48hrs | 1,2 |
| High Temperature Operation | Endurance test applying the electric stress (voltage & current) and the high thermal stress for a long time. | +70°C 48hrs | 2 |
| Low Temperature Operation | Endurance test applying the electric stress (voltage & current) and the low thermal stress for a long time. | -20°C , 48hrs | 1,2 |
| High Temperature / Humidity Operation | Endurance test applying the electric stress (voltage & current) and the high thermal with high humidity stress for a long time. | +40°C , 90% RH , 48hrs | 1,2 |
| Thermal Shock resistance | Endurance test applying the electric stress (voltage & current) during a cycle of low and high thermal stress. | 0°C,30min -> 25°C,5min -> 50°C,30min = 1 cycle<br>10 cycles | |
| Vibration test | Endurance test applying vibration to simulate transportation and use. | 10-55Hz , 15mm amplitude.<br>60 sec in each of 3 directions X,Y,Z<br>For 15 minutes | 3 |
| Static electricity test | Endurance test applying electric static discharge. | VS=800V, RS=1.5kΩ, CS=100pF<br>One time | |

**Note 1:** No condensation to be observed.
**Note 2:** Conducted after 4 hours of storage at 25°C, 0%RH.
**Note 3:** Test performed on product itself, not inside a container.